



ENTERPRISE NETWORK DESIGN PROJECT

Integrated Enterprise Network Architecture

A Practical Guide to High Availability, OSPF Routing & VLAN Security

PROJECT SUPERVISOR

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TRAINING PROGRAM

NTI & ITIDA Initiative

What Are We Building & Why?

The Problem

Single switch setups fail easily.
Unsegmented networks leak private data and slow down under heavy traffic.

Our Solution

We build a 3-tier Cisco architecture connecting HQ, Branch, and Server Farm with zero downtime.

Key Result

Automatic gateway failover, safe departmental isolation, and secured Internet access for all users.

How to Explain This Slide:

"In this project, we designed a complete enterprise network that guarantees continuous operation. If any single router or cable fails, redundant backup paths take over instantly without disconnecting users."

How the 3-Tier Structure Works

Core / Distribution

HQ-MLS1 & MLS2: The core brain. They perform Inter-VLAN routing, OSPF, and gateway failover.

Access Layer

SW4 & SW5: Connects end-user PCs, IP Phones, and local wireless access points directly.

Perimeter Edge

EDGE-R Router: Guards the internal network boundary and translates traffic to the ISP Cloud.

Practical Insight:

By separating the network into Core, Access, and Edge layers, we can easily add new offices or departments in the future without disrupting existing connections.

Why We Divide Devices into VLANs



VLAN ID	Department	Subnet CIDR	Virtual Gateway (HSRP VIP)	Purpose & Security Role
VLAN 10	HR_Admin	10.10.0.64 /27	10.10.0.65	Isolates administrative files & HR PCs
VLAN 20	IT_Staff	10.10.0.96 /27	10.10.0.97	Grants network admin tool access
VLAN 30	Wireless_Users	10.10.0.0 /26	10.10.0.1	Handles Wi-Fi laptops & mobile clients
VLAN 40	Server_Farm	10.10.0.160 /29	10.10.0.161	Secures centralized DHCP, DNS & Web servers
VLAN 50	Voice_VoIP	10.10.0.128 /28	10.10.0.129	Prioritizes voice calls to eliminate delay

 Presentation Tip:

"VLANs stop unnecessary broadcast noise. A computer in HR (VLAN 10) cannot eavesdrop on Server Farm traffic (VLAN 40) unless explicitly allowed by core routing policies."

HSRP: How We Prevent Network Down-Time

HQ-MLS1 (Active)

Priority: 110 (Preempt On)

Handles 100% of normal traffic forwarding for all LAN clients.

HQ-MLS2 (Standby)

Priority: 100 (Backup)

Monitors MLS1 continuously. Takes over in <1 second if MLS1 fails.

Shared Virtual IP

Single Virtual Gateway

End devices only point to VIP. They never notice if physical hardware switches.

Simple Analogy:

Think of HSRP like a twin-engine plane. If Engine 1 (MLS1) shuts down, Engine 2 (MLS2) immediately holds the full load so the plane never drops.

OSPF Area 0: Automatic Path Selection

HQ-MLS1**Router ID:** 1.1.1.1

Main backbone node for HQ subnets.

HQ-MLS2**Router ID:** 2.2.2.2

Redundant backbone node in Area 0.

Branch-MLS1**Router ID:** 3.3.3.3

Connects remote branch users seamlessly.

EDGE-R**Router ID:** 5.5.5.5

Advertises default Internet routes to all routers.

Why OSPF instead of Static Routes?

OSPF routers talk to each other constantly. If a link between HQ and Branch is damaged, OSPF automatically finds an alternative path in milliseconds without human intervention.

LACP EtherChannel: Combining Cable Bandwidth

Port-Channel 1 (Core Backbone)

Switches: HQ-MLS1 <--> HQ-MLS2

Physical Links: Gig1/2/1 + Gig1/2/2 bundled together.

Doubles backbone connection speed to **2 Gbps full-duplex**.

Port-Channel 2 (Access Links)

Switches: Access-SW4 <--> Access-SW5

Physical Links: FastEthernet Fa0/3 + Fa0/4 bundled.

Prevents Spanning Tree from blocking redundant inter-switch links.

How to Explain This Slide:

"Instead of using 1 physical wire that might get congested, LACP combines two wires into one logical super-highway, doubling speed and providing cable failover."



DHCP Server

10.10.0.162

Hands out IP addresses automatically to user PCs via **IP Helper** commands on switches.



DNS Server

10.10.0.163

Translates user domain names into internal IP addresses instantly across all branches.



Web Server

10.10.0.164

Hosts central intranet applications accessible securely by authorized staff members.

How Clients Get IP Addresses:

Even though clients are in different VLANs, the Core Multilayer Switch forwards dynamic IP requests across VLAN boundaries directly to the central DHCP Server.

Internet Access via PAT (Port Address Translation)

Port Address Translation (PAT)

Configuration Concept:

```
ip nat inside source list 1 interface Gig0/0/2 overload
```

Translates hundreds of internal private IPs (10.10.x.x) into **1 public ISP IP**.

Default Gateway Route

Static Default Gateway:

```
ip route 0.0.0.0 0.0.0.0 10.10.2.34
```

Directs any unknown outbound web request straight to the Internet Service Provider.

Why PAT is Essential:

Private IP addresses (10.10.x.x) are forbidden on the public Internet. PAT acts as a security shield, hiding internal IP addresses while letting everyone browse external websites.

 Rapid-PVST

Blocks switching loops instantly while providing fast 2-second link convergence.

 STP PortFast

Bypasses standard 30-second STP delays so employee PCs get connected instantly upon plugging in.

 Voice VLAN 50

Isolates IP phone traffic and prioritizes voice packets to prevent stutter during calls.

 Technical Value:

PortFast saves user time when booting up PCs, while Voice VLAN ensures clear phone calls even when employees download large files at the same time.

✓ Testing Metrics & Proof of Success

100%

Ping Success Rate

Verified complete reachability across all VLANs and Internet.

< 1s

Failover Delay

Sub-second HSRP switchover during simulated primary switch failure.

0

Loop Incidents

Zero switching loops guaranteed by Rapid-PVST & LACP trunking.

💬 Final Verification Summary:

"We tested unplugging the primary MLS switch during active pings. Traffic switched to the standby unit in less than a second with zero packet loss, proving true enterprise reliability."



Thank You! Any Questions?

Enterprise Network Architecture Project — Implemented & Verified

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